

**Proposed Data Centre
Development at No. 2-10,
Tai Yuen Street, Kwai
Chung, N.T.**

Technical Specification for
Formwork & Falsework

Revision: 0

2019-11-18

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1 General

- a) The Contractor shall be responsible for the design, stability, supply, fixing and alignment of all formwork and supports.
- b) The strutting of formwork shall be so arranged that it can be removed without undue shock to the concrete.
- c) Formwork shall be constructed of timber or other approved material. It shall be so designed and constructed as to remain rigid without undue deflection during the placing and compaction of concrete and shall be watertight to prevent loss of grout or mortar from the wet concrete.
- d) Beam and slab soffits shall be supported by vertical props carried down to construction which is sufficiently strong to afford the necessary support without injury to any portion of the permanent structure. No prop directly or indirectly supporting any new construction shall be disturbed for a period of not less than 48 hours after the pouring of the new construction. De propping and re propping to obtain early removal of soffit forms will not normally be permitted.
- e) Prior to commencement of construction the Contractor shall submit and obtain approval for the design of all formwork, method of support of vertical props, and the proposed casting and prop removal sequence.
- f) All stop end formwork at day work joints shall be fixed prior to commencement of concrete placing.

2 Camber in Formwork

- a) Camber shall be provided in the formwork as directed by the Architect or as shown on Drawings. These cambers and any cambers shown on Drawings relate to the formwork immediately before the formwork is struck.

3 Ties for Formwork

- a) All form ties and the position of such ties passing through the formwork shall be to the Architect's approval. Any holes left after removal of ties shall be filled with concrete or mortar of approved composition.

4 Cleaning and Treatment of Formwork

- a) All rubbish, particularly chippings, shavings and sawdust, shall be removed from the interior of all formwork immediately before concrete is placed and the surfaces to be in contact with the concrete shall be cleaned and either thoroughly wetted or treated with a suitable approved mould oil or releasing agent.
- b) Care shall be taken that mould oil or releasing agent is kept out of contact with the reinforcement and does not accumulate at the bottom of formwork. If the formwork is not used within 24 hours, a further coat of mould oil or releasing agent shall be applied before fixing into position.
- c) Mould oil or releasing agent shall be of a type which will have no adverse effect on any finishes to be applied to the concrete nor on the appearance of fair faced concrete, and shall be applied strictly in accordance with the manufacturer's instructions.

5 Removal of Formwork s

- a) The Contractor is responsible for the safe removal of all formwork without exerting shock or vibration to the concrete.
- b) The minimum period for removal of formwork after concreting in normal weather shall be as follows:

- | | | |
|------|--|---------|
| i. | Sides of beams, walls & columns (unloaded) | 1 day |
| ii. | Soffits of slabs (props left under) | 3 days |
| iii. | Props to slabs (unloaded) | 7 days |
| iv. | Soffits of beams (props left under) | 7 days |
| v. | Props to beams (unloaded) | 14 days |
| vi. | Props to cantilevers (unloaded) | 28 days |
- c) Satisfactory test cube results shall be submitted to the Architect prior to removal of props. No props to slabs and beam soffits shall be removed before the concrete has attained its specified 75 percent of the specified 28 day strength.
- d) Formwork may not be removed earlier than the above times without the written approval of the Architect which will only be given with full substantiation that earlier removal of formwork or props will not impair the strength and deflections of the structures.
- e) The Contractor shall be responsible for any damage caused by early or careless removal of formwork.

6 Propping and Reshoring

- a) Where construction loads will exceed the design loads or cause deflections which may affect the tolerance specified for subsequent work, sufficient props shall be left in place to distribute the loads over a number of storeys as directed by the Architect. The Contractor shall follow any instructions as may be given by the Architect on depropping and repropping procedures to the completed structures to ensure that effective load sharing between successive floor will be achieved.
- b) In multistorey construction, particularly where long spans or cantilevers occur, propping to successive floors shall be left in place and carried down far enough to distribute the construction loads to avoid overstress or excessive deflection.
- c) During the stripping and reshoring operation, no construction loads shall be placed on the new construction. The operation shall be planned and performed so that no large areas are left unsupported at any time. The shores replaced shall be positioned over the shores on the floor below.

7 Formwork for Waffle Slab or Ribbed Floor

- a) Moulds for waffle slab or ribbed floor shall be of a type approved by the Architect.

8 Permanent Voids in Concrete

- a) Where permanent voids are to be formed in the concrete, the void formers shall be maintained rigidly in position. The Contractor shall submit to the Architect for his approval details of the measures he proposes to take to ensure accurate formation of the voids.

9 Formwork for Fair Faced Concrete

- a) Smooth fair faced finish to concrete shall be obtained with formwork constructed of, or lined with, wrought timbers, metal or other approved material. The irregularities in the finish shall be no greater than those obtained from the use of wrought thicknessed square edged boards arranged in a uniform pattern. The finish is intended to be left as strut but imperfections such as fins and surface discolouration shall, if required, be made good by methods approved by the Architect.

The formwork shall be made up in the pattern shown on the Drawings. Where no special pattern is shown, the joints in the forms shall be arranged in a neat, regular, and consistent pattern; and the proposed pattern shall be submitted to the Architect for approval.

- b) Form ties shall be arranged in a neat, regular and consistent pattern throughout the work.
- c) Re use of old formwork for fair faced concrete shall be subject to the approval of the Architect.

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